QUESTION PAPER

Model Paper — 03

Mathematics I

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Time: 2 hours **Total marks:** 100

Index number: .....

Important:

- Answer **all** questions on this question paper itself.
- Write your index number correctly in the space provided above.
- Use only the space provided under each question for your answer and for showing how you obtained it.
- Show the relevant steps and the correct units when answering.
- Marks: **Part A** — 2 marks for each question; **Part B** — 10 marks for each question.
- Scrap paper may be used for rough calculations. Calculators are not allowed.

Part	A (1 – 25)	B1	B2	B3	B4	B5	Total
Marks	/50	/10	/10	/10	/10	/10	/100

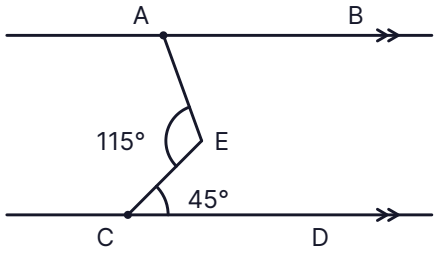
Part A

Answer all questions on this paper itself. (2 marks each — total 50 marks) Take π = 22/7 where needed.

1. At a tea estate, 12 workers can pluck all the tea leaves of a section in 15 days. All the workers work at the same rate. How many more workers are needed to finish the same work in 10 days?

Answer: .....

2. In the figure, AB ∥ CD. If ∠AEC = 115° and ∠ECD = 45°, find the size of ∠BAE.



Answer: .....

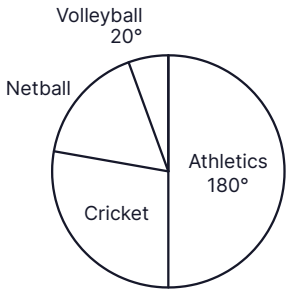
3. Simplify: $\frac{3}{4x} + \frac{5}{6x} - \frac{2}{3x}$

Answer: .....

4. The base of a right prism is a triangle. Put a tick (✓) in the box against the quadrilateral that can **NOT** be a face of this prism.
(i) a square (ii) a rectangle which is not a square (iii) a parallelogram which has no right angle ☐ ☐ ☐

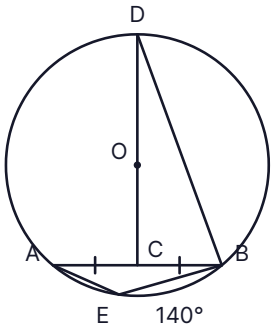
Answer: .....

5. The pie chart shows the sport each student of a school chose for the sports meet. The number of students who chose Volleyball is 8. Find the number of students who chose Athletics.



Answer: .....

6. In the figure, O is the centre of the circle. DOC is a straight line which meets the chord AB at its midpoint C. If $A\hat{E}B = 140°$, find the size of $C\hat{D}B$.



Answer: .....

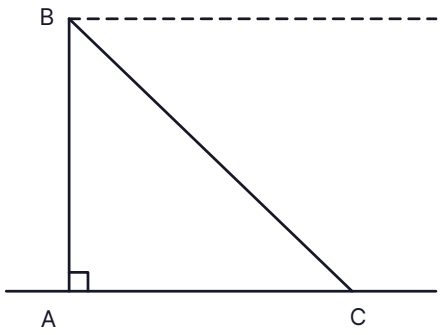
7. Factorise: $2x^2 - 72$

Answer: .....

8. Given that $10^{0.7782} = 6$, write down the value of lg 6000.

Answer: .....

9. In the figure, AB is a vertical flagpole standing on level ground, with the flag at B. Malith stands at C on the ground and looks at the flag. On the figure, mark the angle of elevation of B from C as a, and the angle of depression of C from B as b. Write the relationship between a and b.



Answer: .....

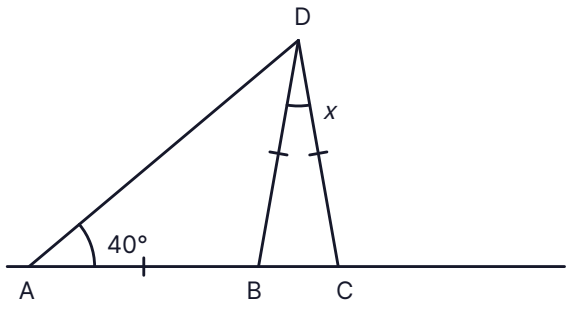
10. The height of a right circular cylinder is 3 times the radius of its base. Find the ratio of the area of paper needed to cover exactly its curved surface to the area of paper needed to cover exactly its base. (Curved surface area of a cylinder = $2\pi rh$)

Answer: .....

11. At a temple festival game, Supun throws two fair tetrahedral (four-faced) dice together. The faces of each die are numbered 2, 3, 5 and 7. Find the probability that the sum of the numbers on the two bottom faces is 10.

Answer: .....

12. In the figure, ABC is a straight line and AB = BD = DC. If ∠DAB = 40°, find the value of x.



Answer: .....

13. Let ξ = {x : x is an integer, $1 \leq x \leq 10$}, A = {multiples of 3} and B = {even numbers}. Put a tick (✓) against the correct statement.
(i) $A \cap B = \{3, 6\}$ (ii) $A \cap B = \{6\}$ (iii) $n(A \cup B) = 8$ ☐ ☐ ☐

Answer: .....

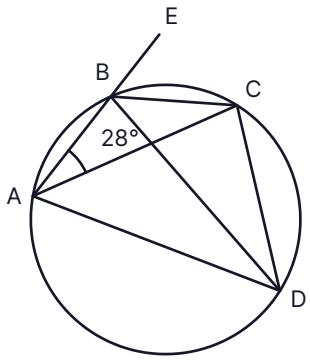
14. Simplify: $6x^2y^3 \times 4x^4y \div 8xy^2$

Answer: .....

15. The second term and the third term of a geometric progression are −4 and −12 respectively. Find the fifth term.

Answer: .....

16. In the figure, ABCD is a cyclic quadrilateral and BD bisects ∠ADC. AB is produced to E. If ∠BAC = 28°, find the size of ∠CBE.



Answer: .....

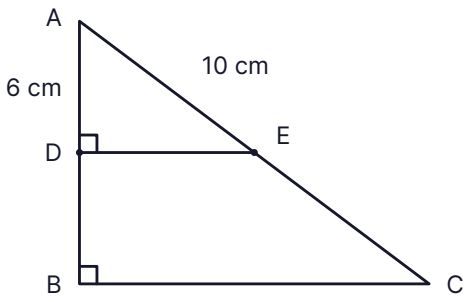
17. Find the highest common factor of $8a^2b^3$, $12a^3b$ and $20ab^2c$.

Answer: .....

18. The area of the triangular base of a right prism is 38.5 cm^2. A solid right circular cylinder has the same height and the same volume as the prism. Find the radius r of the cylinder. (Volume of a cylinder = $\pi r^2 h$)

Answer: .....

19. In triangle ABC, D and E are the midpoints of AB and AC respectively. ∠ADE = 90°, AD = 6 cm and AE = 10 cm. Find the perimeter of the trapezium BCED.

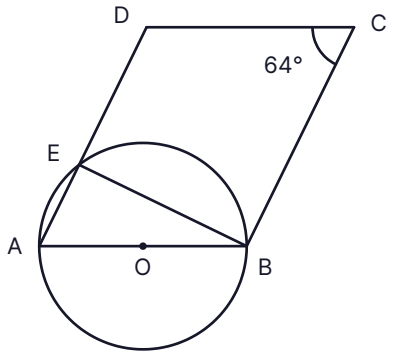


Answer: .....

20. Find the equation of the straight line which passes through the points (2, 4) and (5, 13).

Answer: .....

21. In the figure, O is the centre of the circle and AB is a diameter. ABCD is a parallelogram and the side AD meets the circle at E. If ∠BCD = 64°, find the size of ∠ABE.

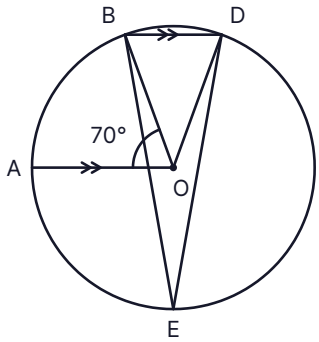


Answer: .....

22. Hasini invested a sum of money for 2 years at an annual compound interest rate of 20%. The interest she earned in the second year was Rs 3 600, and the total amount she had at the end of 2 years was Rs 21 600. Find the sum of money she invested.

Answer: .....

23. In the figure, O is the centre of the circle and OA ∥ BD. If AÔB = 70°, find the size of BÊD.

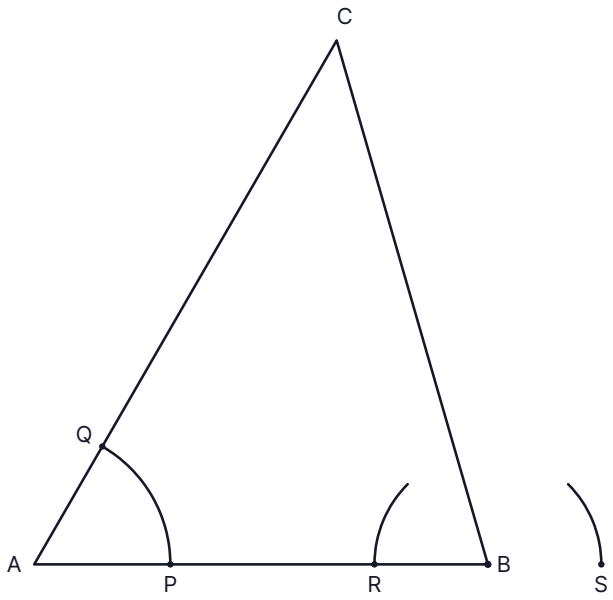


Answer: .....

24. Solve the inequality $5x + 28 > 3$, and write down the negative integers that satisfy it.

Answer: .....

25. In the triangle ABC, a circle touches the side AB at B and also touches the side AC. A part of the construction is already shown. Complete the construction with a ruler and a compass, using what you know, and mark the centre of the circle as O. (You need not draw the circle.)



Answer: .....

Part B

Answer all questions on this paper itself. (10 marks each — total 50 marks) Take π = 22/7 where needed.

1. Hasini, the manager of a tea estate, pays income tax at 10% on the part of her monthly salary that exceeds Rs 100 000. After paying the income tax, she spends $\frac{1}{5}$ of the money she receives on food and $\frac{3}{8}$ of the remainder on other expenses.

(i) What fraction of the money she receives remains after spending on food? (1 mark)

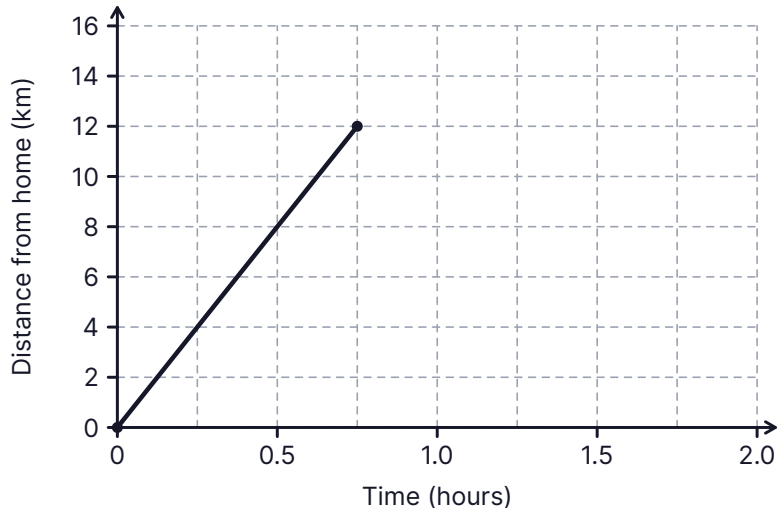
(ii) What fraction of the money she receives remains after spending on both food and other expenses? (2 marks)

(iii) The money that finally remains is Rs 95 000. Find, separately, the money spent on food and the money spent on other expenses. (2 marks)

(iv) Find Hasini's monthly salary before income tax. (3 marks)

(v) Suppose she did not have to pay income tax and the money spent on food remained the same. Express the money spent on food as a percentage of her salary. (2 marks)

2. (a) Supun cycles from his home to the temple festival grounds at a uniform speed. The graph drawn on the grid below shows this part of his journey: he reached the festival grounds, 12 km from his home, after 45 minutes. He stayed there for 30 minutes and then returned home along the same route at a uniform speed, taking 30 minutes.



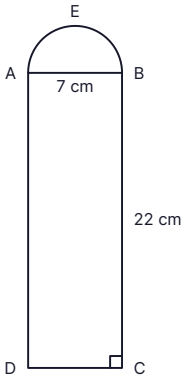
(i) Complete the distance-time graph for the rest of his journey. (3 marks)

(ii) Find his speed on the return journey in km per hour. (2 marks)

(b) The figure shows a plane figure formed by joining a semicircle AEB of diameter 7 cm to a rectangle ABCD, where BC = 22 cm. Take $\pi = 22/7$.

(i) Find the perimeter of the figure. (3 marks)

(ii) The area of the rectangle is how many times the area of the semicircle? (2 marks)



3. Ishara rents out a shop for Rs 9 500 per month and collects the rent for the whole year at once. The Municipal Council has assessed the annual value of the shop as Rs 88 000 and charges annual rates of 20% of the assessed annual value. 10% of the annual rent is spent on maintenance of the shop. Ishara adds some more money to the amount remaining after paying the rates and the maintenance, and buys shares of market value Rs 50 each. Each share pays an annual dividend of Rs 3.50. After one year, he receives a total dividend of Rs 7 000.

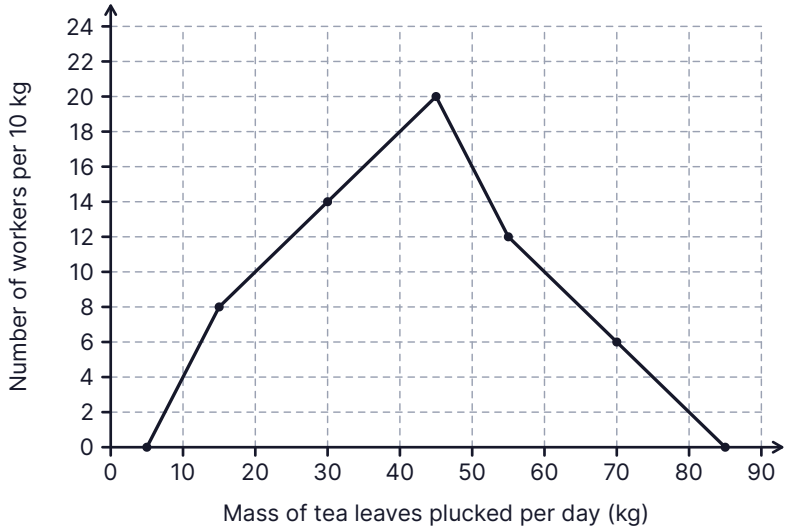
(i) Find the annual rates payable. (2 marks)

(ii) Find the money remaining after paying the rates and the maintenance cost. (2 marks)

(iii) (a) Find the number of shares he bought and the total amount he invested in shares. (3 marks)

(b) The extra money he added is what percentage of the total amount invested? (3 marks)

4. The frequency polygon shows the masses of tea leaves plucked in one day by 80 workers of a tea estate. The classes do not have equal widths: the height of each plotted point is the number of workers **per 10 kg** of class width (the height of the histogram bar for that class). The table below is incomplete.



Class (kg)	Number of workers	Cumulative frequency
10 – 20	8	8
20 – 40	.....	.....
40 – 50	.....	.....
50 – 60	.....	.....
60 – 80	.....	.....

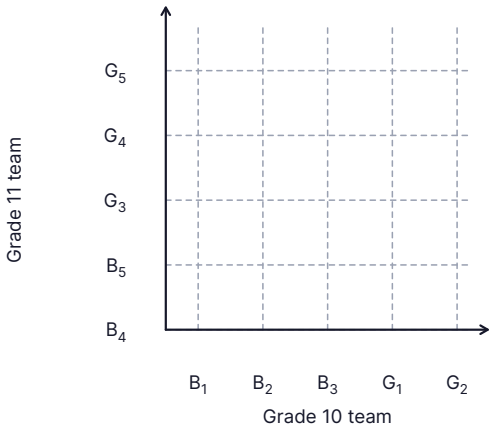
(i) Complete the column “Number of workers” of the table. (3 marks)

(ii) On the same grid, draw the histogram for the data. (3 marks)

(iii) Complete the column “Cumulative frequency” of the table. (2 marks)

(iv) One worker is chosen at random. Find the probability that this worker plucked 50 kg or more. (2 marks)

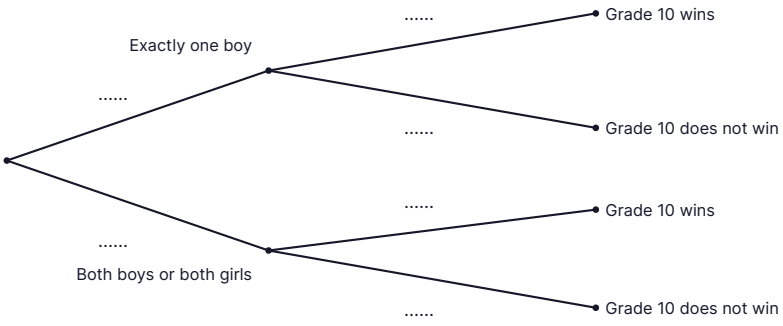
5. For the first round of a relay at a school sports meet, one student is chosen at random from the Grade 10 team and one from the Grade 11 team. The Grade 10 team has three boys (B_1, B_2, B_3) and two girls (G_1, G_2). The Grade 11 team has two boys (B_4, B_5) and three girls (G_3, G_4, G_5). The grid below is for the sample space of the selection of the two students.



(i) Mark the sample space on the grid using the sign "×". (2 marks)

(ii) Enclose with closed curves the outcomes in which exactly one boy is selected, and find the probability of this event. (3 marks)

(iii) If exactly one boy is selected, the probability that the Grade 10 team wins the round is $\frac{3}{5}$; if both are boys or both are girls, this probability is $\frac{1}{2}$. Complete the probabilities on the tree diagram below. (3 marks)



(iv) Find the probability that the Grade 10 team does not win the round. (2 marks)

*** All rights reserved ***